

Impact of Thermal Induced Resonant Frequency Shift on Resonant Inverters

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parameters influences mainly the soft-switching region,

Abstract—Resonant inverters have gained much attention in the area of high frequency power electronics applications due to their high efficiency performances. However, the heart of the resonant converter is the LC tank circuit which significantly depends on the temperature and thus influences converter performance. This paper mainly focuses on the thermal effects on the resonant characteristics followed by its effect on the converter efficiency, which has not been reported anywhere else. This is quantified by incorporating temperature dependent mathematical models for both the inductor and the capacitor analyzing the resultant shift in the resonant frequency, thereby impacting the soft switching and correspondingly the efficiency. Simulations carried on MATLAB and LT spice demonstrates that temperature has adverse effects on the resonant tuning and significant reduction in power transfer efficiency, highlighting the awareness in thermal designing regarding high power density resonant inverters.

Index Terms—Resonant tank, LLC, ZVS, power transfer efficiency and resonant frequency

I. INTRODUCTION

Resonant converters as depicted in Fig.1 fall under the category of specialized DC-DC power converters capable of delivering energy through oscillations rather than abrupt switching transitions. The current or voltage oscillations depends on the series or parallel organization of the resonant tank circuit. Resonant tank circuit that acts as the heart of the resonant converter employs a combination of inductors as well as capacitors. The resonant operation is organized in such a way that the power devices can be turned on or off at the zero crossing of the current or voltage waveform. This reduces the switching stress as well electromagnetic interference making resonant converters extremely suitable for high-frequency and high-power density applications [1]-[2].

Among different resonant topologies, series resonant converters and their derivatives have been widely investigated for isolated high-frequency power conversion. Their inherent current-controlled nature with simple tank configuration and concept of zero current switching (ZCS) switching capability makes them suitable for applications involving reduced turn-off losses and low electromagnetic interference (EMI) are the critical design considerations [3]. Further analysis on resonant converters has been carried out on detailed gain characteristics, region of operation and design constraints which demonstrates, how changes in the resonant tank

voltage transfer ratio, voltage transfer ratio and transient behaviour of the converter [4].

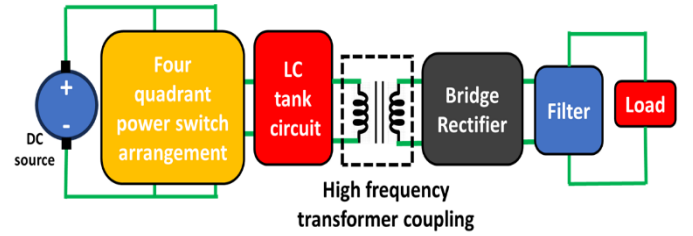


Fig. 1. Series LLC type resonant inverter topology

Due to ample number of advantages, resonant converters have found extensive application in the field of high-frequency power supplies, induction heating, wireless power transfer and high-power DC-DC conversion stages, as far as efficiency along with thermal reliabilities are concerned [5]. Recent developments include focusing on extending the operating range of resonant converters by optimizing the tank design and modulation techniques depicting that, selection of the resonant frequency strongly influences circulating current, efficiency, and power transfer capability [6].

Most resonant converter designs consider fixed component values; but in practice, the inductance and capacitances of the tank circuit are strongly affected by operating temperature. Experimental investigations have proved that even a slight variation in temperature influences both magnetic and dielectric properties, of the inductor and capacitor respectively. [7]. Such thermal induced deviations in the tank parameters can disturb the ZCS condition, increase device current stress as well as degrade efficiency. The high frequency resonant converters operating close to their resonant frequencies are more prone to such effects [8]-[9].

Based on the above discussions, this paper aims to study the influence of thermal dependent variations in resonant tank parameters on resonant frequency, gain characteristics and soft-switching behavior in series LLC resonant converters. The main motive of this work is to aid the design of thermally robust and high-efficiency converters.

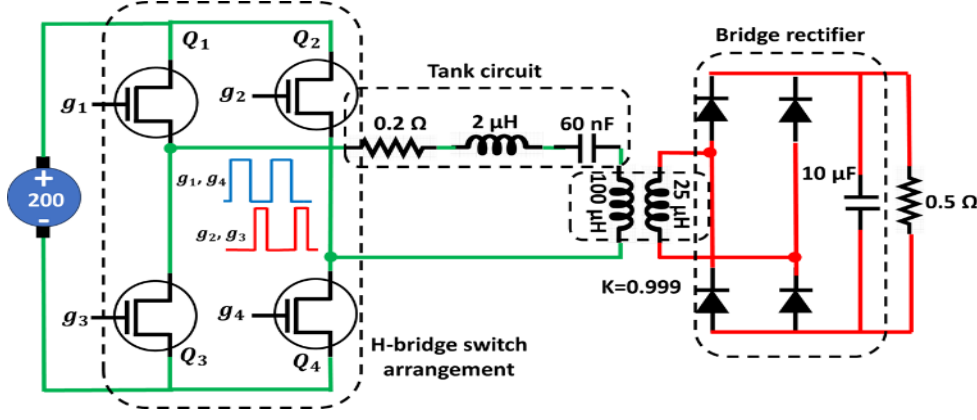


Fig. 2. Full bridge resonant inverter with LLC resonant tank circuit

II. THERMAL MODELLING OF RESONANT TANK CIRCUIT

Fig.2 represents a series LLC resonant converter whose tank circuit bears an initial inductance and capacitance of $2\mu\text{H}$ and 60nF respectively. As stated by Suh *et.al* [10] ferrite cores build the basic block for power inductors and are hence sensitive to the changes in ambient temperature. Similarly, Bun *et.al* [11] came up with similar conclusion while working with epoxy coated polypropylene capacitors.

A. Temperature dependence of ferrite core material

According to [10] the magnetic permeability and temperature follow an inverse relationship. The reported data including variation of relative permeability (μ_r) with temperature (T) has been illustrated in Fig.3(a) and Table I respectively. In reference to Table.1 μ_r can be curve fitted and expressed as

$$\mu_r(T) = aT^3 + bT^2 + cT + d \quad (1)$$

where a, b, c and d are the polynomial constants with values 0.008979, -0.9527, 33.51 and 1815 respectively.

TABLE I
Temperature vs permeability
of ferrite core

T (°C)	μ_r
20	1545
40	1737
60	1937
80	2127
100	2171

TABLE II
Temperature vs coefficient of
polypropylene

T (°C)	$ \alpha $
40	
60	230
70	215
95	200
100	250

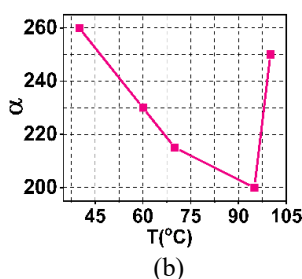
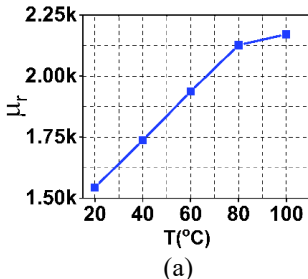


Fig. 3. Thermal variation of (a) Relative permeability of ferrite core (b) thermal coefficient of polypropylene

B. Temperature dependence of polypropylene electrolyte

Polypropylene capacitors are widely used in power electronics converter due to excellent high frequency characteristics and high insulation resistance. Although, manufacturers claim that they are thermally stable but the research findings of Bun *et.al* [11] clearly state that, temperature coefficient of epoxy coated polypropylene is temperature dependent. This has been depicted in Table II and Fig.3(b) respectively and can be curve fitted as

$$\alpha(T) = pT^3 + qT^2 + rT + s \quad (2)$$

where, p, q, r and s are constants given by 0.03697, -3.505, 144.6 and -2441 respectively.

III. THERMAL DEPENDENCE OF RESONANT CHARACTERISTICS

A. Resonant frequency

The resonant frequency f_0 of an LC tank circuit as shown in Fig.2 can be expressed as [12]

$$f_0(T) = \frac{1}{2\pi} \sqrt{\frac{1}{L(T)C(T)} - \frac{R^2}{4L(T)^2}} \quad (3)$$

where, $R, L(T)$ and $C(T)$ are the equivalent series resistance, thermal dependent function of inductance and capacitances respectively. If L_0 indicates inductance at temperature T_0 then

$$L(T) = L_0 \frac{\mu_r(T)}{\mu_r(T_0)} \quad (4)$$

Similarly, the temperature coefficient of capacitance can be expressed as [13]

$$\alpha(T) = \frac{1}{C} \frac{dC(T)}{dT} \quad (5)$$

Integrating both sides, $C(T)$ can be further written as

$$C(T) = C_0 e^{\int_{T_0}^T \alpha(T) dT} \quad (6)$$

The variation of resonant frequency with temperature can be analyzed by substituting Eqn.4 and 6 in Eqn.3. This has been depicted in Fig.4(b). Due to proportionate increase in $L(T)$ and $C(T)$ with temperature described by Eqn.4 and Eqn.6, it is clear from Fig.4(b) that f_0 has almost exponential dependence over change in temperature. Accordingly, the sensitivity variation $\frac{df_0}{dT}$ has been shown by green curve in Fig.4(b). The thermal sensitivity of f_0 has been estimated as $27.332 \text{ Hz}^\circ\text{C}^{-2}$.

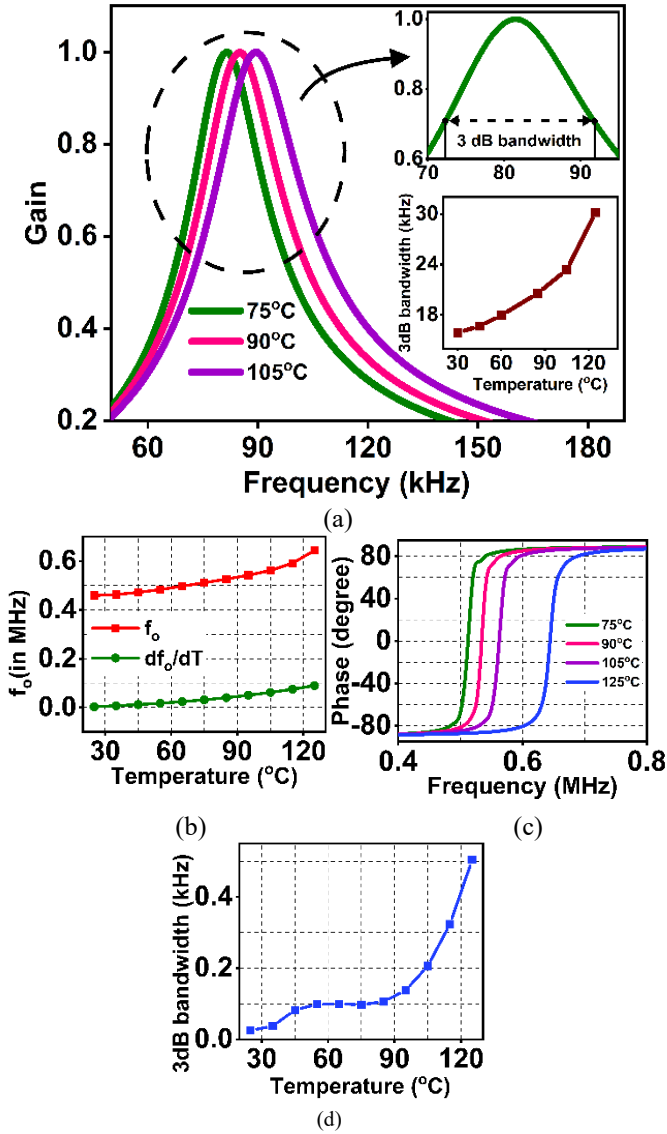


Fig. 4. (a) Thermal variation of gain frequency plot (b) Resonant frequency variation with temperature (c) Phase shift due to temperature (d) Variation of bandwidth sensitivity with temperature

B. Frequency response

The thermal dependent expression of gain $|G(j\omega)|$ in series resonant tank circuit is given by [12]

$$|G(j\omega)| = \frac{1}{\sqrt{1 - \frac{f_0^2(T)L(T)^2}{R^2} \left(\frac{f}{f_0(T)} - \frac{f_0(T)}{f} \right)^2}} \quad (7)$$

where, f is the operating frequency of the tank circuit. Further substituting Eqn.4 and 6 in Eqn.7 the gain frequency plot of the resonant tank can be realized as shown in Fig.4(a). In reference to Fig.4(a) the unity gain frequency shifts right. This is also being confirmed by the phase plot depicted in Fig.4(c). Although, the 3 dB bandwidth increases with increase in temperature, (refer to the inset graph in Fig.4(a)) which appears beneficial at the beginning. However, this is achieved at the cost of increased switching stress, and higher copper losses in resonant inverter discussed in the subsequent section. The thermal bandwidth sensitivity plot shown in Fig.4(d) of the resonant tank circuit signifies bandwidth increment by 14.89 Hz/°C for every 1° rise in temperature beyond 90°C.

IV. SIMULATION ANALYSIS

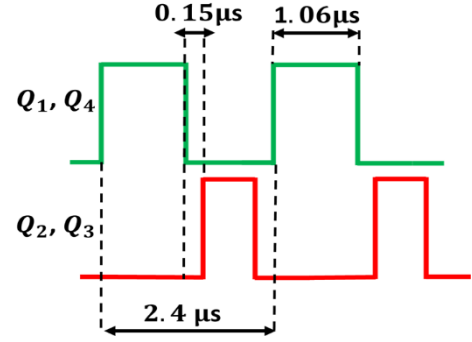


Fig. 5. Switching waveform of the series LLC resonant inverter

TABLE III
Time domain parameters of the switching pulse

Parameters	Value
Time period	2.43 μs
Frequency	410 kHz
Duty ratio	50%
Dead time	150 ns

A. Simulation setup

The analysis of thermal influence on resonant converter has been carried out through simulation in LTspice of full bridge resonant inverter. The simulated circuit illustrated in Fig.1 is a high frequency series LLC resonant inverter supplied by a 200V DC source. Four MOSFET switches are arranged in a full bridge configuration such that, diagonal switch pairs conduct at the same time for generating high frequency bipolar excitation pulses as shown in Fig.5. The time domain parameters have been listed in Table III.

The resonant tank comprises of inductor L_1 and capacitor C_1 whose values are modelled as thermal dependent quantities following Eqn.4 and 6. Furthermore, modelled values have been used to study the resonant frequency shift thereby effecting the converter efficiency. The tank circuit feeds a high frequency transformer that has been considered to be ideal with a very strong coupling coefficient of 0.99.

The secondary side of the transformer, comprises of full bridge rectifier followed by an output filter capacitor of 10 μF and 0.5 Ω resistive load to obtain regulated DC output. The load impedance considering secondary side has been adjusted such that the maximum power transfer theorem is satisfied. The power transformation efficiency has been further analysed under temperature induced parameter drift discussed in subsequent section.

B. Thermal influence on power transfer efficiency

The ability of the resonant converter to effectively transfer power from the tank circuit to the load impedance on the secondary side is mainly governed by the ZVS of the inverter. ZVS condition is further achieved if the condition of the resonance is satisfied. As discussed in section III, since the resonant frequency f_0 varies with temperature, it strongly influences the power transfer capability of the inverter. A shift in f_0 moves the converter away from its optimal resonant

operation, which can violate the ZVS condition and result in elevated current stress on the switching devices. Because the resonant frequency f_0 varies with temperature, it strongly influences the power transfer capability of the inverter. A shift in f_0 moves the converter away from its optimal resonant operation, which can violate the ZVS condition and result in elevated current stress on the switching devices.

The power transfer efficiency η_{PT} of the resonant inverter can be mathematically expressed as

$$\eta_{PT} = \frac{P_R}{P_p} \quad (8)$$

where, P_R is the average power available across load impedance R and P_p is the power available to the primary transformer coil from the inverter. The shifting of resonant frequency (refer Table IV) with temperature leads to the shifting of zero-crossing point of the current waveform. Since the switching intervals of the power switches are fixed, and henceforth the condition of soft switching is significantly affected resulting in poor η_{PT} . The right-shift in the zero-crossing point of the tank current for four different temperatures due to thermal dependent resonant frequency shift has been illustrated in Fig.6.

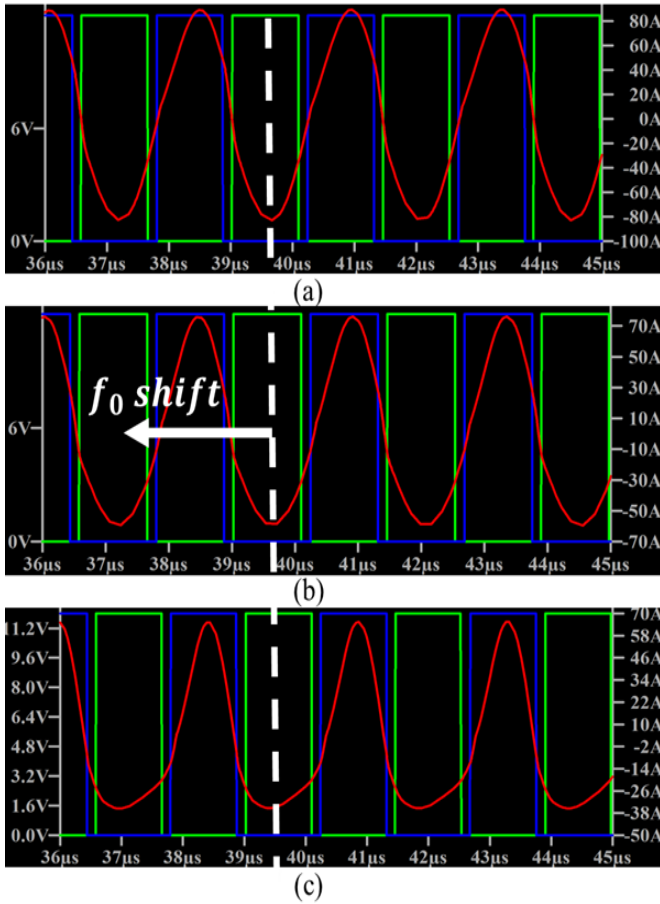


Fig. 6. Spice simulated plots of tank current (red curve) for shift in the resonant frequency at (a) 25°C (b) 75°C (c) 125°C. Here blue and green curves correspond to the gate signals g_1, g_4 and g_2, g_3 respectively.

TABLE IV
Temperature dependent shift in resonant frequency and power transfer efficiency

Temperature (°C)	Resonant frequency (f_0) in kHz	Resonant frequency shift $f_0(25^\circ\text{C}) - f_0(T)$ in kHz	Power Transfer efficiency (η_{PT})
25	459.37	0	91.42 %
30	460.41	1.0411	91.40 %
35	463.01	3.6349	91.37 %
40	466.86	7.493	91.29 %
45	471.74	12.369	91.22 %
50	477.42	18.046	91.08 %
55	483.70	24.325	90.94 %
60	490.40	31.029	90.79 %
65	497.38	38.005	90.78 %
70	504.50	45.132	90.44 %
75	511.71	52.243	90.25 %
80	518.97	59.597	90.05 %
85	526.35	66.974	89.86 %
90	533.99	74.618	89.66 %
95	542.17	82.794	89.46 %
100	551.28	91.911	89.25 %
105	561.93	102.256	89.01 %
110	574.96	115.59	88.74 %
115	591.58	132.21	88.39 %
120	613.60	154.23	87.92 %
125	643.88	184.51	87.21 %

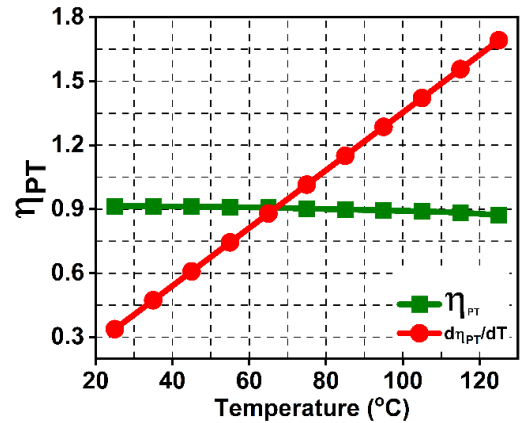


Fig. 7. Power transfer efficiency and sensitivity variation with temperature

The simulated power transfer efficiency η_{PT} values have been listed in Table IV. Correspondingly, the curve fitted plot of η_{PT} values is illustrated in Fig.7, which clearly states that η_{PT} and temperature follows an inverse parabolic relationship given by

$$\eta_{PT} = qT^2 + rT + s \quad (9)$$

where, q , r and s are the weighted constants given by - 0.003484, 0.01354 and 0.9129 respectively. Furthermore, the sensitivity of power transfer efficiency $\frac{d\eta_{PT}}{dT}$ with respect to temperature follows a negative gradient (refer Fig.7) which concludes the fact that thermal rise leads to degradation in power transmission performance. Sensitivity of power transfer efficiency changes by a factor of -0.044 % °C⁻¹ for every 1° rise in temperature. Hence, this becomes extremely crucial for the design engineers to incorporate thermal tuning strategies for reliable power transfer in resonant converters.

V. CONCLUSION

In this paper a detailed analysis of the thermal influence on the power transfer efficiency of the resonant converter has been systematically analyzed. Special focus has been given on how the shift in the resonant frequency due to thermal dependence of inductance and capacitance in the tank circuit affect the converter behaviour. Since resonant tank circuit acts as the heart of the resonant inverter and hence any parameter drift plays a dominant role in the inverter performance. The following points of conclusion are drawn:

- Both ferrite core power inductors as well as polypropylene epoxy coated capacitors exhibit negative temperature coefficient which results in upward shift of resonant frequency with temperature. From the analysis carried out in section III, the thermal sensitivity of resonant frequency varies 27.332 Hz°C⁻¹ for every 1° rise in temperature.
- The right shift in the resonant frequency explained in section IV also leads to noticeable change in the 3dB bandwidth of the resonant tank. Although the sensitivity increases by 14.89 Hz/°C² but this occurs at the cost of increased copper loss, elevated switching stress and deteriorates power transfer efficiency of the resonant inverter.
- Simulation results of full bridge resonant inverter carried out in LTspice demonstrates that, the power transfer efficiency deteriorates with increase in temperature. The resonant frequency shift modifies the zero-crossing point of the current waveform and since the switching frequency is kept constant ZVS is no longer maintained leading to additional stress over the switching devices.
- The power transfer efficiency shows a negative thermal gradient such that the sensitivity of efficiency varies -0.044 % °C⁻¹ for every 1° rise in temperature.

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